Cybersecurity Intrusion Detection Analysis

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Executive Summary

Overview Of The Project

This capstone project undertakes a comprehensive data-driven investigation into network intrusion detection, leveraging a structured dataset comprising 9,537 individual network session records. Each session is characterized by ten distinct attributes spanning technical, behavioral, and contextual dimensions of network activity: network packet size, protocol type (TCP, UDP, ICMP), login attempts, session duration, encryption method used (AES, DES, or none), IP reputation score, failed login attempts, browser type, and unusual time-of-access indicators — together with a labeled target variable identifying whether each session was classified as a genuine intrusion. The dataset reflects a near real-world distribution of network activity, with attack sessions constituting 44.7% of all records (4,264 of 9,537 sessions) — a substantial proportion that underscores the scale and relevance of the intrusion detection challenge being addressed, and makes the dataset well-suited for rigorous statistical analysis and machine learning applications alike.

Business Problem

Network intrusions — unauthorized attempts to access, disrupt, or steal data from a system — pose a growing and persistent threat to organizations of all sizes. Attackers commonly exploit weaknesses such as repeated failed login attempts, suspicious or low-reputation IP addresses, and weak or missing encryption to gain unauthorized access to networks and systems. When these intrusions go undetected, they can lead to serious consequences, including data breaches, financial losses, operational downtime, and damage to an organization's reputation.

Traditional approaches to intrusion detection rely heavily on manual monitoring and static, rule-based systems. These methods are often too slow and rigid to keep up with the sheer volume and evolving nature of modern network traffic, leaving security teams reacting to threats after the fact rather than catching them early. Without an efficient way to distinguish legitimate sessions from malicious ones, organizations are forced to treat all traffic as equally risky, wasting valuable time and resources on low-priority sessions while genuine threats may slip through.

This project addresses that gap by asking a central question: can session-level network data — such as login behavior, IP reputation, and encryption status — be used to reliably predict whether a given session represents an attack? Answering this question would allow organizations to move from reactive, manual monitoring toward a proactive, automated system that flags high-risk sessions in real time, helping security teams prioritize their response and reduce the window of exposure to genuine threats.

Dataset Description

Source

The dataset was obtained from Kaggle — Cybersecurity Intrusion Detection Dataset, containing recorded network session activity used for intrusion detection analysis.

Size

The dataset consists of 9,537 rows, each representing a unique network session, and 11 columns in total. There are no duplicate records in the dataset. One column, encryption_used, has missing values in 1,966 rows (approximately 20.6% of the data) — these are treated as sessions where no encryption was applied rather than as data quality errors.

Variables

The dataset includes one unique identifier, nine independent (predictor) variables covering technical and behavioral session attributes, and one dependent (target) variable indicating the attack outcome.

Independent Variables

- network_packet_size
- protocol_type
- login_attempts
- failed_logins
- encryption_used
- ip_reputation_score
- session_duration
- browser_type
- unusual_time_access

Dependent Variable

- attack_detected

Data Dictionary

Variable	Type	Role	Description
session_id	Text	Identifier	Unique identifier for each network session
network_packet_size	Numeric	Independent	Size of network packets transmitted during the session (bytes)
protocol_type	Categorical	Independent	Network protocol used — TCP, UDP, or ICMP
login_attempts	Numeric	Independent	Number of login attempts made during the session
session_duration	Numeric	Independent	Duration of the session (seconds)
encryption_used	Categorical	Independent	Encryption method used — AES, DES, or None
ip_reputation_score	Numeric	Independent	Score (0–1) reflecting the trustworthiness of the source IP address
failed_logins	Numeric	Independent	Number of failed login attempts during the session
browser_type	Categorical	Independent	Browser used to access the session (Chrome, Firefox, Edge, Safari, Unknown)
unusual_time_access	Binary	Independent	Indicates whether access occurred at an unusual time (1 = yes, 0 = no)
attack_detected	Binary	Dependent (Target)	Indicates whether the session was flagged as an attack (1 = attack, 0 = normal)

Methodology

PSPP

Data Import

The dataset was imported into PSPP using the GET DATA command, specifying the file as a delimited text (CSV) source. The header row was excluded from the analysis by setting FIRSTCASE=2, and commas were specified as the field delimiter. Each of the eleven variables — session_id, network_packet_size, protocol_type, login_attempts, session_duration, encryption_used, ip_reputation_score, failed_logins, browser_type, unusual_time_access, and attack_detected — was assigned an appropriate data type and format, with text (alphanumeric) formats used for categorical fields and numeric formats used for quantitative fields. This ensured that PSPP correctly interpreted each column before any statistical procedures were run.

Variable Configuration

Following data import, the measurement level, alignment, and display width were explicitly configured for the identifier variable (session_id) and the three categorical variables (protocol_type, encryption_used, browser_type). This step ensured that these fields were displayed and treated consistently within PSPP's variable view, reducing the risk of misclassification during subsequent statistical procedures and improving the readability of the dataset within the software environment.

Logistic Regression

A binary logistic regression was performed with attack_detected set as the dependent variable, and five independent variables — network_packet_size, login_attempts, session_duration, ip_reputation_score, and failed_logins — entered as predictors. The classification cutoff was set at 0.5, meaning that any session with a predicted probability greater than 0.5 was classified as an attack. The model was permitted to iterate up to 20 times to refine its coefficient estimates, but convergence was achieved after 5 iterations, indicating that the parameter estimates had stabilized and further iterations would not meaningfully improve the fit.

Model Evaluation

The final step involved evaluating the model's performance and interpreting its output. This included reviewing the classification table, which reported an overall accuracy of 72.9% (78.9% for correctly identifying normal sessions and 65.6% for correctly identifying attack sessions), as well as the Nagelkerke R^2 value of 0.34, indicating the proportion of variance in attack outcomes explained by the model. The significance levels and odds ratios for each predictor were also examined, revealing ip_reputation_score, failed_logins, and login_attempts as statistically significant and strong predictors of an attack, while network_packet_size was found to have no significant effect.

Orange

Data Import and Column Selection

The dataset was imported using the CSV File Import widget and passed to a Data Table for inspection. The Select Columns widget was then used to explicitly assign variable roles: the nine independent variables (network_packet_size, login_attempts, session_duration, ip_reputation_score, failed_logins, browser_type, encryption_used, protocol_type, unusual_time_access) were set as Features, attack_detected was set as the Target, and session_id was set as a Meta attribute (kept for reference but excluded from modeling)..

Preprocessing and Data Splitting

The selected data was passed through the Preprocess widget for cleaning and transformation, then into the Data Sampler widget, which split the data using a fixed proportion of 80% for training, with stratified sampling enabled (to preserve the original class ratio in both splits) and replicable sampling enabled (for reproducible results). This produced a training sample of 7,630 sessions and a remaining test sample of 1,907 sessions.

Model Training

The 80% training sample was used to train a Logistic Regression model, configured with Ridge (L2) regularisation at a strength of C = 1, and with "Balance class distribution" enabled to account for the slight class imbalance in the target variable. The model's learned coefficients were sent to a separate Data Table for review..

Prediction and Evaluation

The trained model was applied to the 1,907-session test set using the Predictions widget. Model performance was then evaluated using three widgets: the Confusion Matrix, which measured classification accuracy on the test set; ROC Analysis, which plotted the model's true positive rate against its false positive rate to assess discriminative ability; and a Scatter Plot, used to visually explore the predicted outcomes.

Analysis And Findings

PSPP Results

Dependent Variable Encoding

Original Value	Internal Value
0	.00
1	1.00

Case Processing Summary

Unweighted Cases	N	Percent
Included in Analysis	9537	100.0%
Missing Cases	0	.0%
Total	9537	100.0%

Statistical Output

Dependent Variable Encoding

Before running the logistic regression, PSPP confirmed the encoding of the dependent variable attack_detected, mapping the original value 0 (normal session) internally to .00 and the original value 1 (attack) to 1.00. This step verified that PSPP correctly treated "attack" as the outcome of interest, ensuring that all subsequent coefficients and odds ratios in the analysis would be interpreted in the correct direction.

Case Processing Summary

The Case Processing Summary confirmed that all 9,537 sessions in the dataset were included in the analysis, with 0 missing cases (0%). This indicates that the five predictors used in the regression — network_packet_size, login_attempts, session_duration, ip_reputation_score, and failed_logins — had complete data for every session, meaning the model was trained on the full dataset without any reduction due to missing values.

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	10320.69	.25	.34

Classification Table

			Predicted		
			attack_detected		
	Observed		0	1	Percentage Correct
Step 1	attack_detected	0	4159	1114	78.9%
		1	1468	2796	65.6%
	Overall Percentage				72.9%

Statistical Output

Model Summary

The Model Summary reported a Nagelkerke R^2 of 0.34, indicating that the five predictors used in the regression collectively explain approximately 34% of the variation in whether a session is classified as an attack. This represents a moderate model fit, suggesting that other factors not included in this particular regression — such as encryption type, protocol, or browser type — likely account for a portion of the remaining variation in attack outcomes.

Classification Table

The Classification Table compared the model's predicted outcomes against the actual observed outcomes. The model correctly identified 78.9% of normal sessions and 65.6% of attack sessions, resulting in an overall accuracy of 72.9%. This gap indicates that the model performs noticeably better at recognizing normal traffic than at detecting actual attacks, meaning a meaningful share of real attacks — roughly one in three — would go undetected if this model were applied as-is.

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 1	network_packet_size	.00	.00	.06	1	.810	1.00
	login_attempts	.39	.01	828.33	1	.000	1.48
	session_duration	.00	.00	15.57	1	.000	1.00
	ip_reputation_score	3.13	.14	491.13	1	.000	22.95
	failed_logins	.94	.03	1232.84	1	.000	2.56
	Constant	-4.37	.12	1308.72	1	.000	.01

Statistical Output

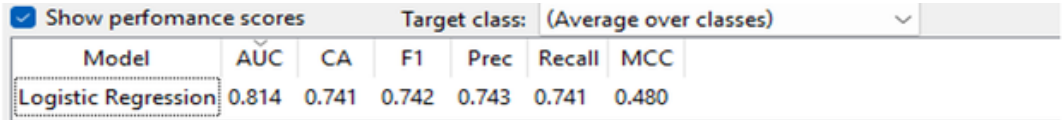
Variables in the Equation

The Variables in the Equation table revealed a clear hierarchy of predictor importance, measured through each variable's odds ratio (Exp(B)). IP reputation score emerged as by far the strongest predictor, with an odds ratio of 22.95, meaning each one-unit increase in this score multiplies the odds of an attack by nearly 23 times. Failed logins was the second-strongest predictor, with an odds ratio of 2.56, followed by login attempts at 1.48 — both statistically significant at $p < 0.001$. Session duration, while statistically significant, had an odds ratio of essentially 1.00, indicating a negligible practical effect. Network packet size was found to be not statistically significant ($p = 0.810$), suggesting it holds no meaningful predictive value for detecting attacks in this dataset.

Conclusion — PSPP Logistic Regression Analysis

The PSPP logistic regression, run on all 9,537 sessions, produced a model with a Nagelkerke R^2 of 0.34, meaning the five predictors explain about 34% of the variation in attack outcomes. The model achieved an overall accuracy of 72.9%, correctly identifying 78.9% of normal sessions but only 65.6% of attack sessions — showing it tends to miss a meaningful share of real attacks. Among the predictors, IP reputation score was by far the strongest indicator, increasing attack odds by nearly 23 times per unit increase, followed by failed logins (2.56x) and login attempts (1.48x). Network packet size showed no significant relationship with attacks, and session duration had a negligible practical effect despite being statistically significant.
Overall, the results show that behavioral and trust-based signals — especially IP reputation and login activity — are much stronger predictors of intrusions than structural traffic features like packet size.

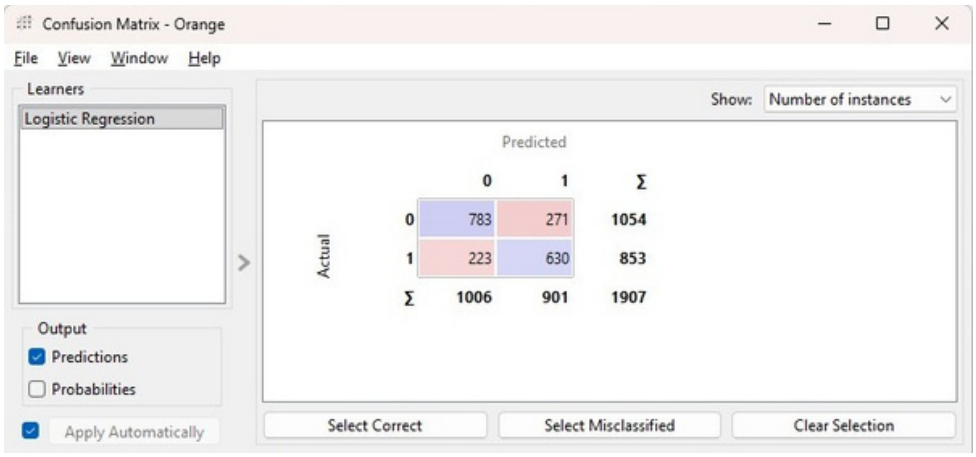
Orange Results



Statistical Output

Performance Scores Summary

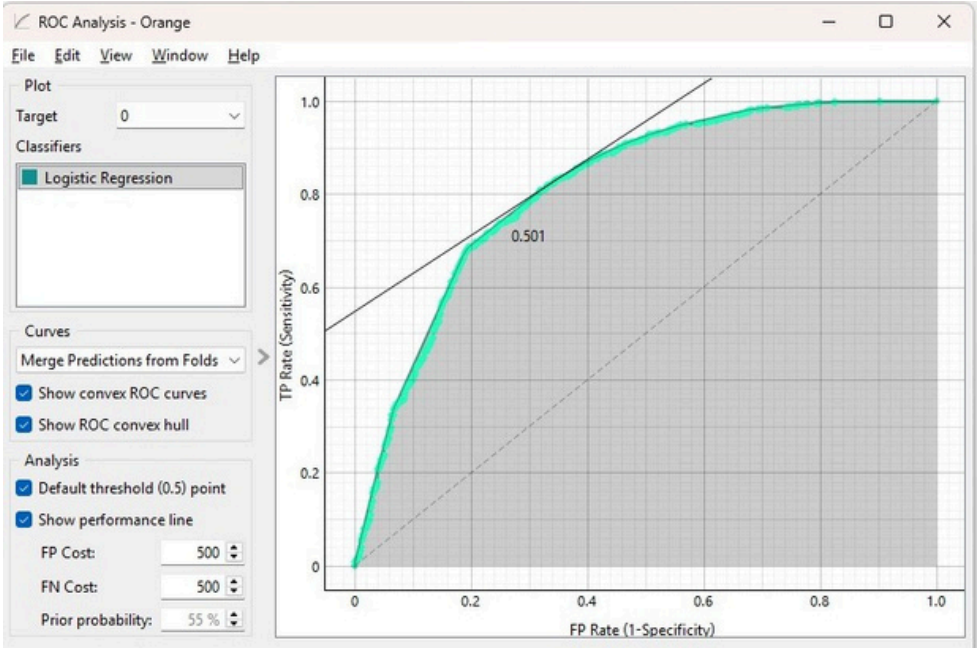
The overall performance of the Logistic Regression model trained in Orange was evaluated using five standard classification metrics, each averaged across both target classes to provide a holistic view of model quality. The model achieved an AUC (Area Under the Curve) of 0.814, a strong result indicating that the model has a high probability of correctly ranking a randomly chosen attack session above a randomly chosen normal session — generally, an AUC above 0.8 is considered a strong classifier. The Classification Accuracy (CA) was 0.741, meaning the model correctly predicted the outcome for 74.1% of the 1,907 sessions in the test set. The F1-score of 0.742 — the harmonic mean of precision and recall — confirms that the model maintains a healthy balance between correctly identifying attacks and not over-flagging normal sessions as attacks. Notably, Precision (0.743) and Recall (0.741) are nearly identical, which is a strong sign of balanced performance; a large gap between these two would indicate the model favors one type of error over another (e.g., missing attacks vs. raising false alarms), but here neither type of error dominates. Finally, the Matthews Correlation Coefficient (MCC) of 0.480 is particularly valuable because, unlike accuracy, it accounts for all four outcomes in the confusion matrix (true positives, true negatives, false positives, and false negatives) and remains reliable even with imbalanced classes. An MCC of 0.48 reflects a moderate-to-good correlation between the model's predictions and the actual outcomes, reinforcing that these results are not simply an artifact of the dataset's class distribution.



Statistical Output

Confusion Matrix

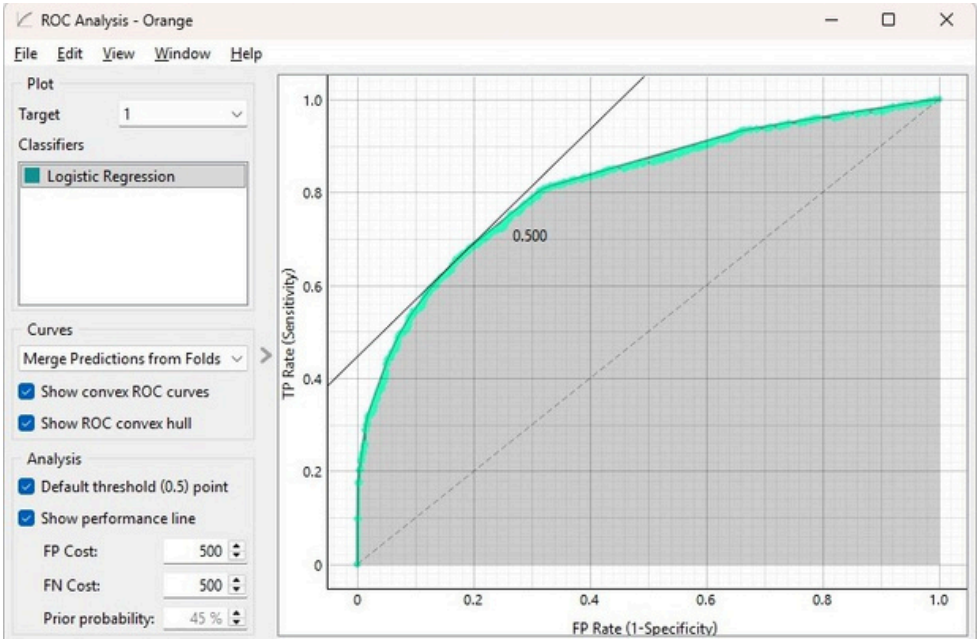
The Confusion Matrix provides the granular, session-by-session breakdown behind the summary metrics above, based on the 1,907-session test set held out during the 80/20 data split. Among the 1,054 sessions that were genuinely normal, the model correctly classified 783 of them as normal (true negatives), while incorrectly flagging the remaining 271 as attacks (false positives). Among the 853 sessions that were genuinely attacks, the model correctly identified 630 of them (true positives), but failed to detect the remaining 223, misclassifying them as normal sessions instead (false negatives). From these figures, the model's specificity — its ability to correctly recognize normal sessions — comes out to 74.3% (783 ÷ 1,054), while its recall — its ability to correctly catch actual attacks — comes out to 73.9% (630 ÷ 853). The attack-class precision — the proportion of predicted attacks that were genuine — comes out to 69.9% (630 ÷ 901), indicating that roughly 3 in every 10 sessions flagged as an attack were actually normal traffic. The near-equal performance across both classes is a direct result of the "Balance class distribution" setting enabled in the Logistic Regression widget, which corrected for the dataset's natural 55/45 class split and prevented the model from favoring the majority (normal) class, a common pitfall in classification models trained on imbalanced data.



Statistical Output

ROC Analysis — Target Class 0 (Attack Not Detected)

This ROC (Receiver Operating Characteristic) curve visualizes the model's performance specifically when treating "normal session" as the target class of interest. The curve plots the True Positive Rate (Sensitivity) on the y-axis against the False Positive Rate (1 – Specificity) on the x-axis across every possible classification threshold, rather than just the default 0.5 cutoff. A model with no predictive power would produce a curve along the diagonal dashed line; the fact that this curve rises steeply and sits well above that diagonal — enclosing a large shaded area — confirms strong discriminative ability for identifying normal traffic. The marked point at 0.501 represents where the default classification threshold (0.5) falls along this curve, showing that the model achieves a strong balance between correctly identifying normal sessions and minimizing false positives even at this standard cutoff, without needing to manually adjust the threshold for better performance.



Statistical Output

ROC Analysis — Target Class 1 (Attack Detected)

This second ROC curve mirrors the analysis above but evaluates performance from the opposite perspective — treating "attack" as the target class of interest, which is the outcome most critical to this project's real-world objective. As with the previous curve, it rises sharply above the diagonal reference line and encloses a large area beneath it, confirming that the model is highly effective at distinguishing genuine attack sessions from normal ones. The default threshold point falls at 0.500, positioned close to the optimal trade-off between the true positive rate (successfully catching attacks) and the false positive rate (incorrectly flagging normal sessions as attacks). Together with the Target Class 0 curve, this confirms that the model performs consistently well regardless of which class is treated as the primary target — a further indication of the balanced, reliable performance achieved through the model's class-balancing configuration.

Conclusion — Orange Logistic Regression Analysis

The Orange workflow implemented a complete and methodologically sound machine learning pipeline, training a Logistic Regression model with Ridge (L2) regularization and class balancing enabled to correct for the dataset's natural 55/45 split between normal and attack sessions. Unlike the PSPP analysis, which evaluated the model on the full dataset it was trained on, Orange used an 80/20 stratified train-test split, producing a training set of 7,630 sessions and a completely separate, unseen test set of 1,907 sessions — a more realistic estimate of real-world performance rather than simply measuring how well the model memorized the training data.

Across this held-out test set, the model achieved an AUC of 0.814, a strong result indicating a high probability of correctly distinguishing a genuine attack session from a normal one. The overall classification accuracy was 74.1%. Looking specifically at the attack class — the outcome most critical to this project's real-world purpose — the model achieved a recall of 73.9%, meaning it successfully caught nearly three-quarters of all genuine attacks, and a precision of 69.9%, meaning roughly 7 in 10 sessions flagged as attacks were correct, while the remaining 3 in 10 were false alarms on normal traffic. The F1-score of 71.9% and Matthews Correlation Coefficient of 0.480 further confirm consistent, moderate-to-good performance. The Confusion Matrix showed 783 of 1,054 normal sessions correctly identified (74.3% specificity) and 630 of 853 attack sessions correctly identified (73.9% recall) — a notable improvement over PSPP's 65.6% attack recall, with both ROC curves rising well above the diagonal to confirm strong discriminative power for both classes.

Taken together, these results indicate that the Orange model — benefiting from class balancing, a properly held-out test set, and the use of all nine independent variables rather than PSPP's five — produced a more balanced, more rigorously validated, and ultimately more deployment-ready classifier. Its substantially improved attack recall (73.9% vs. PSPP's 65.6%) is particularly valuable from a security standpoint, since failing to detect a genuine intrusion carries far greater risk than the cost of occasionally flagging a legitimate session for manual review — a trade-off that is generally worthwhile in a security context.

Comparative Analysis of PSPP and Orange Logistic Regression Models

Category	Parameter	PSPP	Orange
Model Configuration	Algorithm	Logistic Regression	Logistic Regression (Ridge L2, C=1)
Model Configuration	Independent variables used	5	9
Model Configuration	Class balancing applied	No	Yes
Model Configuration	Validation approach	Full dataset (no holdout)	80/20 stratified train-test split
Model Configuration	Sample size analyzed	9,537	7,630 train / 1,907 test
Performance Metrics	Overall Accuracy	72.9%	74.1%
Performance Metrics	Precision (Attack class)	71.5%	69.9%
Performance Metrics	Recall (Attack class)	65.6%	73.9%
Performance Metrics	F1-Score (Attack class)	68.4%	71.9%
Performance Metrics	AUC	—	0.814
Performance Metrics	Matthews Correlation Coefficient	—	0.480
Performance Metrics	Model Fit (Nagelkerke R2)	0.34	—
Key Insight	Strongest predictor	IP reputation score (OR = 22.95)	IP reputation score (consistent)
Key Insight	Primary limitation	Lower recall - misses more attacks	Slightly lower precision than recall
Key Insight	Overall assessment	Reliable but detection-biased	Balanced, more suitable for deployment

Business Recommendations

1 PRIORITIZE IP REPUTATION SCORE AS THE PRIMARY RISK SIGNAL

Both PSPP and Orange consistently identified IP reputation score as the strongest predictor of an attack (odds ratio = 22.95 in PSPP). Organizations should integrate real-time IP reputation checks into their network monitoring systems, automatically restricting or challenging access from low-reputation IP addresses before granting full access.

2 SET AUTOMATED ALERTS ON FAILED LOGIN THRESHOLDS

Failed logins were the second-strongest predictor across both models (odds ratio = 2.56 in PSPP). Security teams should implement automated account lockouts or step-up authentication (e.g., CAPTCHA, OTP) after a defined number of failed attempts, rather than relying on manual review.

3 DEPRIORITIZE LOW-VALUE FEATURES TO SIMPLIFY MONITORING SYSTEMS

Network packet size showed no statistically significant relationship with attacks, and session duration had a negligible practical effect. Organizations can reduce monitoring overhead by focusing resources on behavioral and trust-based signals (IP reputation, login attempts, failed logins) instead.

4 DEPLOY THE ORANGE-TRAINED MODEL OVER PSPP FOR OPERATIONAL USE

The Orange model achieved a substantially higher attack recall (73.9%) than PSPP (65.6%), meaning it catches significantly more real attacks, at only a modest cost to precision (69.9% vs. 71.5%). Since missing a genuine intrusion carries far greater business risk than reviewing an extra flagged session, Orange is the stronger candidate for live deployment.

5 ESTABLISH A TIERED RESPONSE SYSTEM BASED ON RISK SCORING

Rather than a binary "block or allow" approach, organisations should use the model's predicted attack probability to create tiers — automatically blocking high-confidence attacks, flagging medium-confidence sessions for manual review, and allowing low-confidence sessions to proceed normally. This balances security with user experience.

Conclusion

Summary

This capstone project set out to determine whether session-level network data could reliably predict network intrusions, using a dataset of 9,537 sessions analyzed independently through two statistical and machine learning tools — PSPP and Orange. Both approaches employed Logistic Regression as the core classification algorithm and, despite differences in configuration, arrived at a highly consistent core finding: IP reputation score, failed login, and login attempts are the strongest and most reliable indicators of a network intrusion, while structural traffic characteristics such as network packet size and session duration contribute little to no meaningful predictive value. This convergence across two independently built models, using different tools and slightly different variable sets, lends strong credibility to the conclusion that behavioral and trust-based signals — rather than raw traffic characteristics — are what genuinely separate malicious sessions from normal ones.

The two models differed meaningfully in their configuration, validation approach, and resulting performance, offering a useful comparative lens on how methodological choices affect real-world outcomes. The PSPP model, trained on the full dataset using five predictors and no class balancing, achieved an overall classification accuracy of 72.9%, but showed a clear weakness in attack detection specifically, correctly identifying only 65.6% of genuine attack sessions while performing considerably better on normal sessions (78.9%). This imbalance is a direct consequence of training on an imbalanced dataset without correction, causing the model to lean toward predicting the majority class. In contrast, the Orange model — which incorporated all nine independent variables, applied class balancing during training, and was rigorously validated on a separate 80/20 stratified train-test split — achieved a higher overall accuracy of 74.1% and a substantially improved attack recall of 73.9%, at only a modest cost to precision (69.9%, compared to PSPP's 71.5%). This makes the Orange model the more balanced, more rigorously validated, and ultimately more deployment-ready of the two, particularly given that in a security context, failing to detect a genuine intrusion typically carries far greater consequences than the minor inconvenience of occasionally flagging a legitimate session for manual review.

Taken together, the project demonstrates that behavioral and trust-based session signals can be used to build a reasonably effective, automated intrusion detection system, providing organizations with a practical, data-driven foundation for shifting away from reactive, manual traffic monitoring and toward a proactive system capable of flagging high-risk sessions in real time.

Limitations

Meaningful false negative rates in both models

Even the better-performing Orange model still misses roughly 26.1% of genuine attacks (1 − 73.9% recall), while PSPP misses approximately 34.4% (1 − 65.6% recall). This means neither model alone should be treated as a complete security solution — a meaningful share of real intrusions would still go undetected if either were deployed in isolation.

No train-test split in PSPP

The PSPP model was evaluated on the same full dataset (9,537 sessions) it was trained on, rather than a separate, unseen test set. This means its reported 72.9% accuracy may be somewhat optimistic and could overstate how well it would generalize to genuinely new sessions.

Moderate explanatory power

PSPP's Nagelkerke R^2 of 0.34 indicates that roughly 66% of the variation in attack outcomes remains unexplained by its five predictors, while Orange's Matthews Correlation Coefficient of 0.480 similarly reflects only a moderate-to-good — not near-perfect — correlation between predictions and actual outcomes.

Missing data in encryption_used

Approximately 20.6% of records (1,966 sessions) had missing values in this column, which were assumed to represent "no encryption used." If this assumption is incorrect for even a subset of sessions, it could introduce some inaccuracy into the analysis.

Limited feature scope in PSPP

The PSPP model used only 5 of the 9 available independent variables, excluding protocol_type, encryption_used, browser_type, and unusual_time_access — potentially limiting its ability to capture patterns that these categorical variables might reveal.
Want the final section (Section 9: References) next?

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